Gradient-Guided Multiscale Focal Attention Network for Remote Sensing Scene Classification

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***Abstract*— Remote sensing scene classification (RSSC) aims to understand and analyze the semantic information at the scene level with complex geographical properties. Despite the profound success of advanced deep models in automatically capturing hierarchical embedding representations and the gradual dominant trend in RSSC, it still remains a great challenge to precisely focus on targets at variable scales that are considered highly relevant to the corresponding scene and separated from the background. Motivated by this recognition, in this article, we present the gradient-guided multiscale focal attention network (GMFANet) for RSSC to adaptively localize the representative multiscale semantic representation for complex scenes. In particular, a lightweight parameterized hierarchical multiscale attention (HMA) mechanism is proposed, which constitutes the main aim of adaptively enhancing physical detail and high-level semantic information at different layers, rather than regarding each scale set with equivalent insight, while eliminating redundant information inherent in conventional attention mechanisms. Subsequently, a gradient-guided spatial focused attention (GSFA) module is specifically designed to accurately localize critical regions at multiple scales, with the dynamic combination of gradient-activated reference attention map and prediction attention map from supervised information-based learning. In addition, a curriculum-driven dynamic attention fusion (CDAF) strategy is tailored to fuse the spatial attention above from easy to hard for avoiding from poor local optimum and decreasing the early learning ambiguity. Our extensive comparative experiments and ablation analyses implemented on real-world public RSSC datasets indicate that our approach achieves the state-of-the-art performance exactly. The code is available at https://github.com/bling2beyond/GMFANet.**

***Index Terms*— Attention mechanism, curriculum learning, multiscale learning, remote sensing, scene classification.**

I. Introduction

As THE rapid development of sensor and Earth observation technologies, high-resolution remote sensing images have become increasingly accessible, and massive Earth observation data enable intelligent interpretation for remote sensing gradually feasible. The high spatial resolution enables remote sensing images to cover a considerable number of diverse object categories, which in turn necessitates the exploration of global contextual information so as to understand and discriminate the scenes contained in the image [1]. The remote sensing scene classification (RSSC), which understands and analyzes the target properties and semantic information sufficiently to allocate scene-aware label to the remote sensing imagery, has served as a dominant research line along fields such as urban monitoring, environmental assessment, and land migration [2], [3], [4], [5], by virtue of the strong geographical feature description capability. In contrast to natural image classification, remote sensing imagery is characterized by a more complex background and significant variance of ground target scales, and the coexistence of multiple target classes still poses challenges for scene discrimination. These are prone to both large intraclass variance and interclass similarity [1].

In order to extract features relevant to the scene semantics for describing remote sensing imagery, early research focuses on hand-designed descriptors with invariance to globally or locally represent the raw data space. Texture feature as representative target information in remote sensing data is commonly extracted in localized regions of the image in a relative difference analysis. Completed local binary pattern (CLBP) is proposed for land use scenario analysis in [6]. The improved version, multiscale CLBP (MS-CLBP), is also concurrently designed to enhance multiresolution performance. In [7], the CLBP descriptor is utilized to describe texture information with local rotational invariance on multiscale image patches. The color histogram with translation and rotation invariance provides a global description of the image from the viewpoint of the color distribution. In [8], a variety of color histogram-based methods are tested for aerial scenes. For instance, color autocorrelogram (ACC) [9] is capable of encoding spatial color distributions in an image. Border-interior pixel classification (BIC) [10] computes the edge pixel and interior pixel color histograms separately, and even exceeds the accuracy of ConvNets on a given dataset. In addition, local or global low-level descriptors such as GIST [11], scale-invariant feature transform (SIFT) [12], and histogram of oriented gradients (HOG) [13] can be utilized directly for the description of the scene image [8], [14]. However, the hand-crafted low-level scene features are tremendously limited with prior knowledge and cumbersome design [1].

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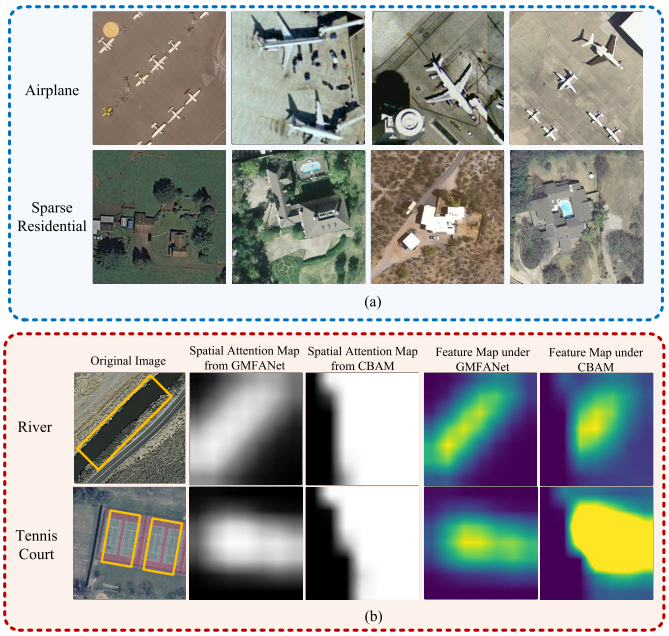


Fig. 1. Some examples to demonstrate (a) challenges in the presence of variable-scale targets on RSSC data and (b) challenges that conventional spatial attention mechanisms such as CBAM [25] fail to focus on critical geographical areas, which are indicated by orange rectangular boxes.

As deep neural networks (DNNs) gradually dominate the area of computer vision [15] and intelligent interpretation of remote sensing [16], learning-based feature extraction technique for RSSC is increasingly favored by researchers. In the beginning, autoencoders were used to perform dimensionality reduction on the raw image space to enhance the distinguishability. Zhang et al. [17] leveraged sparse autoencoder to automatically extract representative deep representations, and subsequently, support vector machine (SVM) was employed to classify the learned features, which is considered as a two-stage solution. Du et al. [18] learned high-level deep embedding representations in an unsupervised manner with stacked convolutional denoising autoencoders as feature extractor, which facilitated the improvement of subsequent classifier performance. Subsequently, convolutional neural network (CNN)-based approaches were successively proposed to simultaneously perform hierarchical semantic feature extraction and scene analysis in an end-to-end manner. He et al. [19] distinctively designed skip connections and covariance pooling to constitute the skip-connected covariance (SCCov) network to tackle the large-scale variance issue in RSSC. Wang et al. [20] proposed a global–local two-stream scheme to extract multiscale representations and locate critical regions. In addition, transformer-based methods emerged to fully exploit the contextual information of the background [21], [22], [23].

Scene analysis for remote sensing imagery still confronts two major challenges, which are depicted in Fig. 1, especially in complex scenarios.

1) The diversity of covered scenario categories, multiple data sources, multiple resolution properties, variable-scale targets, and complex backgrounds increase intraclass diversity. Existing multiscale learning schemes tend to fuse each scale feature equally, ignoring the disparate contributions of different scales to representative deep representations. Attention mechanism provides a novel insight into the possibility of adaptively reallocating the contribution of scale-aware elements to the fused feature. Nevertheless, it has been suggested in the literature that capturing the dependencies of all the elements is both inefficient and unnecessary [24]. It is, thus, essential to develop a lightweight hierarchical scale-aware attention mechanism for efficient adaptive preservation of multiscale information.
2) In RSSC, scene analysis poses a great challenge as the same scene often distributes diverse types of geographical objects, and different scenes may share similar targets. Although the spatial attention mechanism can suppress redundant spatial information, it still fails in RSSC to separate highly relevant foreground targets from background information due to the aforementioned problems. It is imperative to design a focal spatial attention mechanism that localizes critical regions with representative semantic representations at multiple scales.

In this article, we present the gradient-guided multiscale focal attention network (GMFANet) to address the above issues for RSSC. The hierarchical features as extracted by CNNs possess different receptive fields, corresponding to different levels of semantic information. In general, shallow features have small receptive fields and therefore contain more detailed information and represent richness for small-scale targets. Deeper networks have larger receptive fields and enable the extraction of advanced semantic information, whereas detailed information gets lost and is preferable for detecting large-scale targets. Inspired by the mechanism of human vision, we propose a lightweight parameterized hierarchical multiscale attention (HMA) scheme to adaptively fuse multiscale information instead of the existing work that deals with each scale feature set equally, neglecting the distinct contributions of different scale information to the semantic representation. In contrast to considering dependencies between all elements, we only consider the relations at the adjacency hierarchical scale and long-range skip-connected scale form a random sampler, in addition, convert the computation of attention scores into learnable parameters with fewer redundant connections. To compensate for the limitation of spatial attention in scene analysis, we specifically design a gradient-guided spatial focused attention (GSFA) module to spotlight target regions that are highly correlated with the scene, enhancing the separability of the target representation from the background information. In order to address the tendency to fall into local optima and generate ambiguity during optimization for the GSFA module, curriculum-driven dynamic attention fusion (CDAF) strategy, which extends sample-based curriculum learning to fusion and learning of focal attention maps from easy to hard, is specially designed to enhance the generalization of the model.

The contributions of this article can be summarized as follows.

1) We present the GMFANet for RSSC to not only adaptively localize the representative multiscale semantic representation for complex scenes, but also enhance the

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significant detection and separability from the background for physical objects.
2) The HMA scheme, GSFA mechanism, and CDAF strategy are presented for the first time to perform adaptive enhancement of physical details and high-level semantic information, accurately localize critical regions separated from background, and suppress the learning ambiguity during optimization.
3) Numerous experiments, ablation analyses, and visualizations are conducted to substantiate the superiority with the indication that our method performs favorably against state-of-the-art algorithms on benchmark datasets.

The organization of this article is as follows. First, we briefly introduce the relevant content and background in Section II. In Section III, the proposed GMFANet is described in detail. The experiments are shown in Section IV. Finally, Section V contains some conclusions and describes future work.

II. Related Works and Background

A. Remote Sensing Scene Representation Learning

Deep representation learning has become an essential domain for the machine learning community. With the help of layer-by-layer structure and inductive bias of local perception, CNNs can automatically learn more representative and discriminative high-level semantic patterns [26]. As well, it can substitute for handcrafted feature design to learn deep embedded semantic representations for complex scenarios in remote sensing imagery [27], [28]. In the beginning stage, CNNs are simply treated as feature extractors to establish a scene representation for RSSC, and classifiers are subsequently trained on the feature set to determine scene categories [29]. In order to learn efficient image representations, Lu et al. [30] designed a shallow weighted deconvolution network to learn informative feature map sets in an unsupervised form, and the aggregated features were subsequently fed into classification. Chaib et al. [31] employed a pretrained visual geometry group network to extract high-level features from very high resolution (VHR) images before implementing an SVM as a classifier for scene classification. Furthermore, Yuan et al. [32] presented to extract global and local feature representations from deep model pretrained on natural images, and after rearrangement to train a multilayer perceptron completing the classification.

Recent advances in research are dedicated to scene representation as end-to-end learning in pursuit of the state-of-the-art performance [27]. In [27], metric learning regularization was introduced to guide CNNs in learning more discriminative feature representations. Zeng et al. [33] comprehensively considered global-context features and local-object-level features to enhance the capability to characterize dispersed small targets in remote sensing images. In order to enhance the robustness of rotation-variant images, Fang et al. [34] additionally considered frequency-domain information along with spatial-domain information for joint learning. In order to improve the feature extraction capability for scene classification, Chen et al. [4] encoded the contextual information of remote sensing scene images as local features. Despite the tremendous success of existing representation learning methods based on CNNs, it is still a great challenge to accurately focus on variable-scale targets that are highly correlated with the scene and separated from the background. This article intends to explore an adaptive hierarchical multiscale feature fusion scheme and spatial focused attention mechanism to accurately learn representative semantic representations of scenes.

B. Multiscale Learning in RSSC

Multiscale representation learning has dominated the domain of computer vision [35], [36], [37], [38]. Multiscale is the representation of an image at different levels of granularity, with fine-grained representations reflecting more detail and coarse-grained representations providing stronger semantic representation. In RSSC, variable scales often bring great challenges to scene feature extraction as a result of multiresolution sources and abundant object sizes. In addition, for CNN-based approaches, the deep and shallow layers of the model contain diverse information, and the deep layers have a relatively large receptive field and a strong ability to characterize semantic information but a weak ability to characterize geometric information; on the contrary, the shallow layers have a relatively small receptive field and a strong ability to characterize detailed information but a weak ability to characterize semantic information. Parallel multibranching and serially connected structures might be considered as potential saviors for solving multiscale representation learning in RSSC.

For the parallel multibranch structure, the commonly used multiscale learning modules for natural images, such as spatial pyramid pooling [39], [40], pyramid pooling module [41], and atrous spatial pyramid pooling [42], can be directly applicable to RSSC. Furthermore, Hu et al. [43] extracted features from the input image at different scales and encoded the coalescent features, thus constructing a global feature representation. Zheng et al. [44] performed multiscale max-pooling of deep features to improve invariance to multiscale deformations. Wang et al. [20] proposed multiscale fusion of local area with global information with supervised key area detection strategy. In order to develop efficient multiscale representations, the efficient pooling channel attention module, consisting of spatial pyramid max-pooling and spatial pyramid average-pooling, is specifically designed [45].

The skip connection of hierarchical multiscale features allows for the same enhancement of variable-scale geologic object representation. In [46], multilayer features from a pretrained CNN model are integrated for scene analysis. Similarly, He et al. [47] utilized the multilayer features of pretrained CNN to compute the covariance matrix thereby leveraging the multiscale complementary information. In addition, Lu et al. [29] presented a feature aggregation CNN to fully consider different spatial and semantic information. In order to solve large-scale variance in RSSC, He et al. [19] designed skip connections to provide multiresolution information for subsequent covariance pooling. There are still numerous studies that investigate serially connected multiscale feature learning schemes for RSSC [4], [21], [31].

Existing multilayer feature fusion tends to combine features from each scale equally, with ignoring disparate contributions. This article provides an insightful exploration of the

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lightweight parameterized HMA scheme by means of the adaptive reallocation property of the attention mechanism, which in turn provides adaptive enhancement of physical detail and high-level semantic information at different layers.

C. Attention Mechanism in Computer Vision

The attention mechanism is inspired by the concept in cognitive science that humans selectively focus on a portion of all information while ignoring other visible information due to a bottleneck in information processing, which has been extensively practiced in computer vision [48]. Most attention mechanisms can be described in the following form:

$$\mathcal{O} = \Pi(\mathbb{S}(I), \mathbb{M}(I)) \tag{1}$$

where I and $\mathcal{O}$ correspond to the input features and the output information after attention enhancement, respectively. $\mathbb{S}(I)$ denotes the computation of attention scores on the input features for capturing the importance of discriminative elements, and $\mathbb{M}(I)$ can be regarded as a further encoding of the input features, and in common spatial or channel attention, this operation is usually replaced by an identity mapping. In particular, it refers to the learning of value vectors for self-attention mechanism. $\Pi(\mathbb{S}(I), \mathbb{M}(I))$ further adaptively realigns the modified input feature information $\mathbb{M}(I)$ according to the attention score $\mathbb{S}(I)$, with paying more attention to the key regions and eliminating redundant information.

Existing attention mechanisms are more inclined to concentrate on spatial and channel information. Hu et al. [49] designed the squeeze-and-excitation (SE) block to compress spatial information and capture dependencies between channels. On this basis, a variety of improved versions of channel attention mechanism have been successively proposed [50], [51]. Spatial attention then provides adaptive information selection on spatial regions. In [52] and [53], the researchers in different manners pursue for important regions with invariance to geometric transformations. In contrast to the explicit descriptions, [54] and [55] make implicit predictions for the region of interest. Woo et al. [25] combined channel attention with spatial attention to capture both spectral and spatial neighborhood dependencies for CNNs. In addition, self-attention is also an important component of the attention mechanism favoring the capture of global self-information [56], [57], [58]. Similarly, attention mechanisms have been implemented to pay attention to key geographical regions or spectral information on RSSC. Wang et al. [59] proposed attention recurrent convolutional network to discard noncritical information for improving scene classification performance. Chen et al. [4] designed the global context spatial attention module to encode the contextual information of remote sensing scene image into local feature.

Nevertheless, existing attention mechanisms only focus on spatial or channel information. This article intends to exploit the adaptive allocation between scales information by leveraging the characteristics of attention mechanisms. In addition, capturing the dependencies of all elements is both inefficient and unnecessary [24], so it is necessary to develop a lightweight parameterized multiscale attention scheme.

D. Gradient-Guided Activation

Stochastic gradient descent (SGD) combined with backpropagation is a popularly favored deep network optimization algorithm. The well-known Adadelta [60], RMSProp [61], Adam [62], etc., [63] provide an efficient solution to the nonconvex optimization problem of complex deep models with the gradient information to guide the model parameters closer to the optimum. In addition to offering guidance for optimization, the gradient information can also build additional activation for deep models. Gradient information can serve to outline the essential impact of the disharmony of the large difference in the number of positive and negative examples in the one-stage object detectors. The gradient harmonizing mechanism was designed to mitigate this divergence [64]. In [65], the gradient norm assists the optimizer in finding a flat minimum, enhancing the generalization of the deep model. Voynov et al. [66] exploited gradient information to guide diffusion generation. In addition, Selvaraju et al. [67] used the gradient of the target concept injected into the convolutional layer to visualize the important regions of the image for predicting the specific concept. Dual-gradients localization framework was novelly presented to implement weakly supervised precise localization on any convolutional layer [68].

In this article, we attempt to explore the gradient information in a self-pace manner to guide the deep model to focus on key ground object regions at multiple hierarchical scales, which in turn improves the precise understanding of the scene.

III. Methodology

In this section, the proposed GMFANet for scene understanding and semantic analysis is depicted. First, the overview architecture, learning objective, and optimization strategy of our network are presented to provide a holistic view. Subsequently, our proposed HMA mechanism, GSFA module, and CDAF strategy, as the innovative and critical components of our network, will be described individually.

A. Overview and Preliminaries of GMFANet

The framework of GMFANet is shown in Fig. 2. The original remote sensing image is fed into the feature extractor for layer-by-layer feature extraction, and our method is a backbone-agnostic algorithm that can be plugged into most convolutional network backbones, such as VGG [69], GoogleNet [70], and ResNet [71]. The convolutional block in Fig. 2 represents a feature extractor layered by a group of convolutional and nonlinear layers, which varies depending on the backbone employed. It can be expressed as

$$\mathbf{x}^l = \mathcal{F}^l\left(\mathbf{x}^{l-1}; \Theta^l\right) \tag{2}$$

where $\mathbf{x}^l$ denotes the output of the convolutional block of layer l, and $\mathbf{x}^{l-1}$ denotes the input and also the output of the convolutional block of layer $l-1$. $\mathcal{F}^l$ is the feature extractor with parameters Θ^l.

Shallow features preserve more detailed information, which facilitates the analysis of fine-grained targets. Since pooling or convolutional layers with stride enable exponential downsampling on the feature map, which in turn extracts deep

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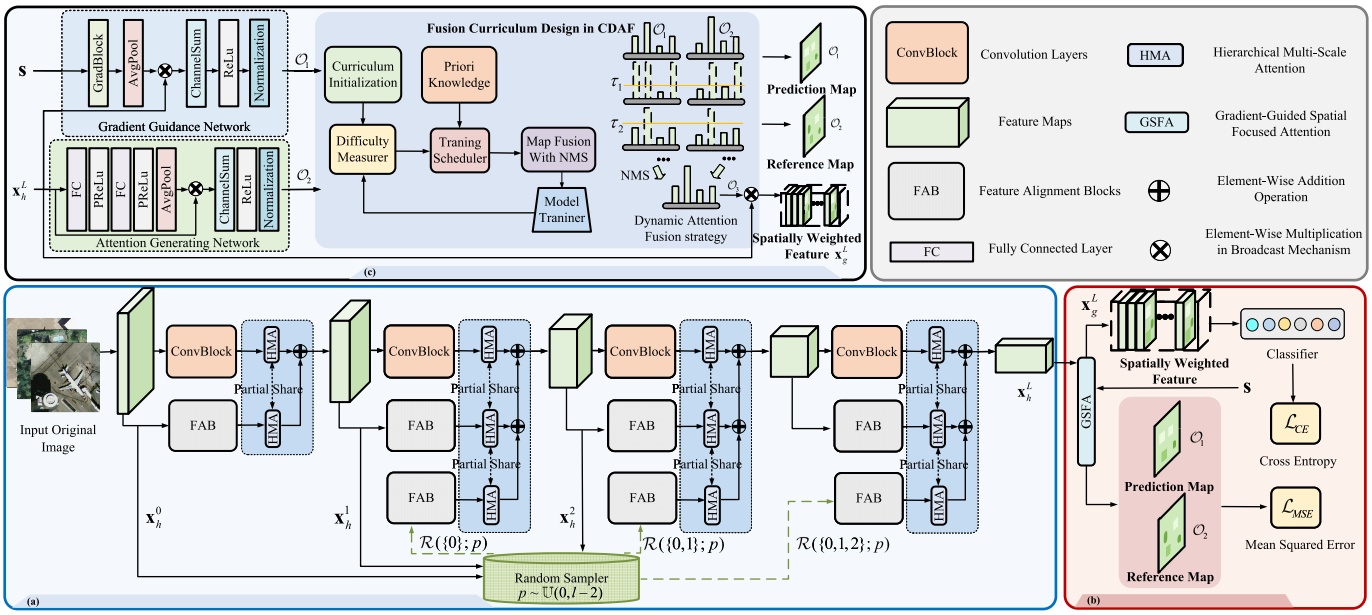


Fig. 2. Framework of the proposed GMFANet for RSSC, which consists of (a) convolutional backbone with HMA mechanism that adaptively fuses multiscale hierarchical features via both long-range skip and short-range connections and (b) scene classification head with feature rectification under GSFA that localizes critical regions separated from the background. In order to demonstrate the mechanism of GSFA even further, the structure and learning strategy of GSFA are purposely enlarged to (c) pipeline of GSFA mechanism with CDAF that suppresses the learning ambiguity during optimization.

semantic information, it is more conducive to the comprehension of large-scale targets. In order to preserve the memory of deep semantic features for shallow detailed features, HMA is proposed to adaptively fuse adjacent hierarchical features. Meanwhile, in order to avoid the sharp drop of features in previous layers except for the adjacent layer, we pick a scale from them for adaptive fusion via a random sampler, and the randomness of long-range skip connections enhances the robustness of the deep model to diverse scales information even more. With the support of HMA, (2) can be evolved as

$$\mathbf{x}^l = \mathcal{F}^l\left(\mathcal{H}\left(\mathbf{x}_h^{l-1}, \Phi_1^{l-1}\left(\mathbf{x}_h^{l-2}\right), \Phi_2^{l-1}\left(\mathbf{x}^r\right); A^{l-1}\right); \Theta^l\right) \tag{3}$$

where $\mathcal{H}$ represents the HMA fusion module utilized for adaptive multiscale enhancement, along with a set of learnable parameters A, and $\mathbf{x}^r$ indicates randomly sampled long-range front layer fused features. In addition, Φ denotes the feature alignment block (FAB) for aligning features from different layers to layer l both in space and channel dimensions. In particular, Φ is defined differently for different layers. The details of HMA fusion will be presented in Section III-B.

For the purpose of overcoming the inability that conventional spatial attention mechanisms fail in focusing on crucial ground object regions in remote sensing image scene analysis, we present a unique GSFA module after the last layer of HMA fusion. The pipeline of GSFA module is described in Fig. 2. It receives two parts of the input, the fused features of the last HMA fusion $\mathbf{x}_h^L$ and the logic vector obtained by the classifier during preinference $\mathbf{s} \in [0, 1]^{\mathcal{C}\times 1}$, where

$$\mathbf{x}_h^L = \mathcal{H}\left(\mathbf{x}^L, \Phi\left(\mathbf{x}_h^{L-1}\right), \Phi\left(\mathbf{x}^r\right); A^L\right). \tag{4}$$

Preinference refers to feeding the current batch of samples into the network for one inference by setting all of the focal attention elements to ones before each forward propagation.

First, the logic vector $\mathbf{s}$ and the multiscale fusion feature $\mathbf{x}_h^L$ are fed into two subnetworks, i.e., the gradient guidance network and the attention generating network, respectively, to obtain the reference map with focal attention $\mathcal{O}_1$ and the prediction attention map $\mathcal{O}_2$. Then, the attention maps $\mathcal{O}_1$ and $\mathcal{O}_2$ are dynamically fused from easy to hard under the proposed CDAF strategy. Finally, the fused attention map is calibrated to $\mathbf{x}_h^L$ to obtain multiscale focal features with target-aware field and the union $\mathbf{x}_g^L$ is fed into the classifier for scene classification

$$\mathbf{x}_g^L = \mathcal{V}(\mathcal{O}_1, \mathcal{O}_2; \tau) \odot \mathbf{x}_h^L, \quad \text{s.t. } \tau \in \Psi \tag{5}$$

where $\mathcal{V}$ is the fusion function with the curriculum injection parameter τ, $\odot$ is the elementwise multiplication in the broadcast mechanism, and Ψ is introduced as a curriculum constraint to control the attention fusion procedure with an easy-to-hard strategy, which in turn drives the model toward a more generalized local optima. Details and analysis of the GSFA module and CDAF strategy will be presented successively in Sections III-C and III-D, respectively.

It is worth noting that the gradient guidance network above is a nonlearning module that activates critical ground regions by calculating gradient information, and the attention generating subnetwork that fits prediction map $\mathcal{O}_2$ with proximity capabilities through supervised learning from the reference map $\mathcal{O}_1$ and classification. Therefore, in general, the learning objective of GMFANet consists of cross-entropy loss ℓ_{CE} on the classifier and supervised learning on the two attention maps, which is accomplished with mean square error (MSE) loss ℓ_{MSE}. Along with dynamically evolving curriculum information, the learning objective can be expressed in the following formula:

$$\mathcal{L} = \ell_{\text{CE}} + \beta \times \ell_{\text{MSE}} \tag{6}$$

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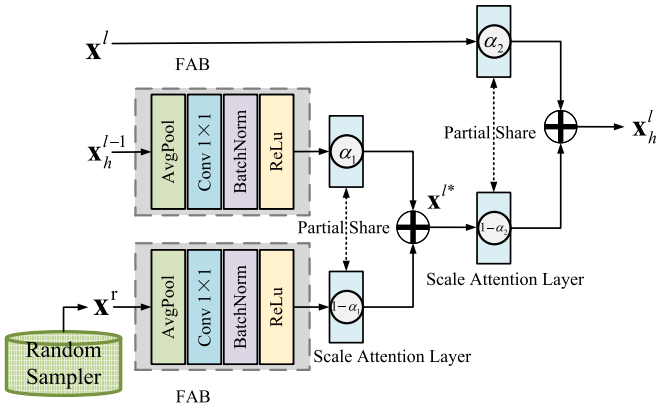


Fig. 3. Framework of the proposed HMA mechanism.

where β is utilized to balance cross-entropy loss and MSE loss.

B. HMA Mechanism

In order to perform adaptive fusion of different hierarchical scale information that enables our model to be robust to varying scale targets, we introduce a unique insight in attention mechanism and specifically design the HMA mechanism, which constitutes the main aim of performing adaptive enhancement of physical detail and high-level semantic information at different layers rather than regarding each scale set with equivalent insight. The straightforward idea would be compressing spatial and channel information at different scales and adaptive enhancement in the scale dimension by means of channel attention mechanism. To the best of authors' knowledge, most existing scale-aware attention mechanisms focus on stacking spatial or channel attention modules at multiple scales [72], [73], [74]. There is an urgent demand to develop an attention scheme for scale-specific features. On the one hand, unlike spatial or channel attention that exploits the output attention scores of a network adapted to the input information, the layer scale information is closely related to the network structure, where the position of different layers determines their filtering treatment for a particular granularity. On the other hand, motivated by the recognition that capturing the dependencies of all the elements is both inefficient and unnecessary [24], this section seeks to explore a lightweight scheme for efficient adaptive preservation of multiscale information. Attention scores are converted into learnable network parameters embedded in the attention layer, recalibrating the scale features in a cascading fashion. The framework of the HMA mechanism is shown in Fig. 3.

The HMA receives three inputs, including the output from the extractor in current layer $\mathbf{x}^l$, the features of the neighboring previous layer $\mathbf{x}_h^{l-1}$, and the long-range skip-connected $\mathbf{x}^r$ from the random sampler. The sampler randomly samples one layer from the previous layers except for $\mathbf{x}_h^{l-1}$, and the sampling probability follows a uniform distribution. The random sampling operation can be described as

$$\mathbf{x}^r = \mathcal{R}(\{0, 1, \ldots, l-2\}; p), \quad p \sim \mathbb{U}(0, l-2) \tag{7}$$

where operation $\mathcal{R}(\{0, 1, \ldots, l-2\}; p)$ denotes random sampling from the index set $\{0, 1, \ldots, l-2\}$ of previous layers before $\mathbf{x}_h^{l-1}$ with probability p and returns the corresponding feature $\mathbf{x}^r \in \{\mathbf{x}_h^0, \mathbf{x}_h^1, \ldots, \mathbf{x}_h^{l-2}\}$. p obeys a uniform distribution $\mathbb{U}$ with bounded parameters 0 and $l-2$. While short-range connections can preserve the information of the previous hierarchical layer from being lost to some extent, this situation gets progressively poorer as the layer gets deeper, and longer range connections for shallow information maintain the memory of the semantic features for detailed information such as texture. Randomness further enhances the robustness of the classifier to interpret information at varying scales. First, $\mathbf{x}_h^{l-1}$ and $\mathbf{x}^r$ pass through the FAB, which consists of average pooling operation, convolutional layer with 1×1 kernel, batch normalization layer, and rectified linear unit (ReLu) nonlinear activation layer, respectively, to align the spatial and channel sizes with $\mathbf{x}^l$. The outputs are denoted as $\Phi_1^l(\mathbf{x}_h^{l-1})$ and $\Phi_2^l(\mathbf{x}^r)$. After a cascaded connection and the recalibration of learnable attention scores

$$\mathbf{x}^{l*} = \alpha_1 \odot \Phi_1^l\left(\mathbf{x}_h^{l-1}\right) + (1-\alpha_1) \odot \Phi_2^l\left(\mathbf{x}^r\right) \tag{8}$$

the output of HMA can be expressed as

$$\mathbf{x}_h^l = \alpha_2 \odot \mathbf{x}^l + (1-\alpha_2) \odot \mathbf{x}^{l*} \tag{9}$$

where α_1 and α_2 are the shared learnable parameters in the scale attention layers, coupled with $1-\alpha_1$ and $1-\alpha_2$ are regarded as attention scores. $\odot$ is the elementwise multiplication in the broadcast mechanism.

C. GSFA Mechanism

Conventional spatial attention mechanisms often fail in remote sensing scene analysis with a lack of ability to focus on critical ground object areas. In this section, we describe the proposed GSFA module. The framework is shown schematically in Fig. 4. The logic vector obtained by the classifier during preinference $\mathbf{s} \in [0, 1]^{\mathcal{C}\times 1}$ is fed forward into a gradient guidance network for generating reference spatial attention scores, and the gradient activation information can spotlight regions that are significant for scene discrimination [67]. The gradient guidance network consists of gradient block, global average pooling layer, channel-dimensional summation operation, ReLu nonlinear activation, and normalization layer. The reference map for the output can be represented as

$$\mathcal{O}_1 = \delta\left(\mathrm{ReLu}\left(\sum_k\left(\mathrm{Avg}\left(\frac{\partial \mathbf{s}^c}{\partial \mathbf{x}_h^L}\right) \odot \mathbf{x}_h^L\right)\right)\right) \tag{10}$$

where $\mathbf{s}^c$ is the logic value for class c, k denotes the number of feature channel, δ is the min–max normalization, and $\mathrm{Avg}(\cdot)$ denotes the global average pooling operation.

The fused features of the last HMA fusion $\mathbf{x}_h^L$ are fed forward into attention generating network to fit a predicted spatial focused attention map $\mathcal{O}_2$ in a learned manner. The attention generating network consists of two fully connected layers, and parametric ReLu (PReLu) nonlinear activation is used to fit the gradient information above; then, the same global average pooling layer, channel-dimensional summation operation, ReLu nonlinear activation, and normalization layer

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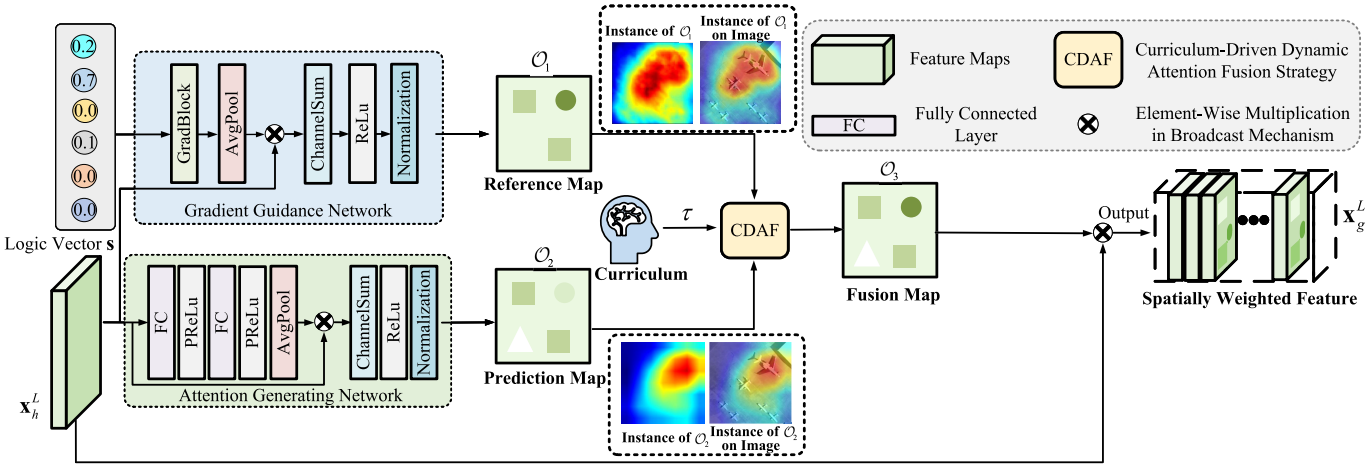


Fig. 4. Framework of the proposed GSFA module.

are arranged. The output is formulated as

$$\mathcal{O}_2 = \delta\left(\mathrm{ReLu}\left(\sum_k \left(\mathrm{Avg}\left(\mathbb{G}^c\right) \odot \mathbf{x}_h^L\right)\right)\right) \tag{11}$$

where

$$\mathbb{G}^c = \mathrm{PReLu}\left(\mathbf{W}_2 \mathrm{PReLu}\left(\mathbf{W}_1 \mathbf{x}_h^L\right)\right). \tag{12}$$

$\mathbb{G}^c$ is the fit gradient information for class c. $\mathbf{W}_1 \in \mathbb{R}^{(k/2)\times k}$ and $\mathbf{W}_2 \in \mathbb{R}^{k\times(k/2)}$ are the parameter matrices of the fully connected layer, respectively. After obtaining the prediction and reference maps, a CDAF strategy is employed to fuse them from easy to hard to obtain the final fused attention map to recalibrate the feature map $\mathbf{x}_h^L$

$$\mathcal{O}_3 = \mathcal{V}(\mathcal{O}_1, \mathcal{O}_2; \tau), \quad \text{s.t. } \tau \in \Psi. \tag{13}$$

Eventually, the output $\mathbf{x}_g^L$ is formulated by (5). In the training phase, $\mathcal{O}_1$ and $\mathcal{O}_2$ are used to construct the MSE loss in (6) for supervising the attention generation network with gradient activation information.

D. CDAF Strategy

There is the literature implicitly indicating that gradient activation has inherent region localization ability [67], [68]. In addition, the attention map is usually unsupervised, which may result in low-importance regions being given higher weights [75], as depicted in Fig. 1. Consequently, facilitated by certain adaptive learnable ability together with focusing ability, prediction map which is supervised by the both reference map with gradient activation and classification labels will provide richer and more flexible spatial attention allocation. We fuse the reference map with the prediction map to obtain the final spatial focused attention. However, simultaneous access to gradient activation map for supervision and generated attention map tends to increase learning ambiguity. In order to reduce the learning ambiguity in the early stages of training and fusing for better convergence of the model, a curriculum learning strategy is employed to fuse the attention from easy to hard. Initially, most curriculum learning methods were based on samples [76]. Subsequently, individual-based [77], node-based [78], and cluster-based [79] curriculum learning were explored. Under this inspiration, we heuristically extend sample-based curriculum learning to fusion and learning of focal attention maps from easy to hard. The framework of our CDAF strategy is shown in Fig. 5.

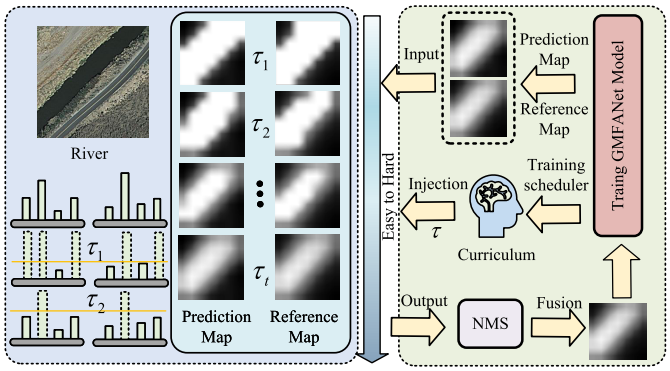


Fig. 5. Framework of the proposed CDAF strategy.

The learning and generation of involved attention maps can be categorized into two stages, region segmentation and score prediction. In this case, region segmentation is considered to be the simpler task that is utilized to facilitate the separability of the target from the background. Attention score fitting and computation is considered as complex task, which requires a more fine-grained representation to reflect the importance of spatial location. A dynamic threshold τ was set to control the attention fusion process from easy to hard according to the prefabricated curriculum Ψ. In the initial phase of model training, CDAF prefers to focus on the separation of the target region from the background, and the curriculum threshold τ should be set to a low takeoff value, with regions greater than this threshold being assigned a hard score, and regions below this threshold retaining their reference or prediction attention scores. This process can be described as

$$\mathcal{O}^\tau = \mathbb{1}(\mathcal{O}_{ij} > \tau), \quad \text{s.t. } \tau \in \Psi \tag{14}$$

where $\mathbb{1}$ indicates that the value is set to 1 for elements that satisfy the condition; otherwise, it remains unchanged.

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Algorithm 1 Learning and Optimization of GMFANet

Input: Training set $\mathbf{X} = \{x_i; y_i\}_{i=1}^{n}$; Hyperparameter τ_0 and β; Maximum iterations: MaxEpoch.

Output: The optimized model GMFANet with network parameters $\theta_G^{MaxEpoch}$.

Initialize network parameters θ_G^0 and initial curriculum τ_0.
for $epoch = 1 \rightarrow MaxEpoch$ **do**
  for *Batch* $\{x_i; y_i\}_{i=1}^{B}$ *in* $\mathbf{X}$ **do**
    Forward:
    Feeding batch samples for preinference to obtain s with setting all of the elements in $\mathcal{O}_3$ to ones.
    Update curriculum information Ψ in CDAF via $\tau \leftarrow$ Eq. (15).
    Feeding batch samples for forward propagation.
    Compute hierarchical feature $\mathbf{x}^l \leftarrow$ Eq. (3) with HMA.
    Compute fused feature of the last layer $\mathbf{x}_h^L \leftarrow$ Eq. (4).
    Compute reference map $\mathcal{O}_1 \leftarrow$ Eq. (10).
    Compute prediction map $\mathcal{O}_2 \leftarrow$ Eq. (11).
    Compute focal attention $\mathcal{O}_3 \leftarrow \mathcal{V}(\mathcal{O}_1, \mathcal{O}_2; \tau)$.
    Recalibrating features $\mathbf{x}_g^L \leftarrow$ Eq. (5).
    Construct the overall loss $\mathcal{L} \leftarrow$ Eq. (6).
    Backward:
    Update parameters θ_G^{epoch} with SGD and backpropagation algorithm.
  end
end

As training advances, the value of τ increases dynamically, and we make it here to satisfy the linear rule

$$\Psi : \tau = \tau_0 + \text{itr} * \Delta\tau. \tag{15}$$

τ_0 is the initial threshold, $\Delta\tau$ is taken as curriculum step, and itr is referred as the number of update iterations. Until τ reaches the maximum value of attention scores, otherwise it increases dynamically according to the above pattern. In addition, nonmaximum suppression (NMS) is selected as a fusion function. NMS finds focal attention scores with high spatial values and remove the lower scoring map between the prediction map and reference map, which will not destroy strongly activated areas

$$\mathcal{V}(\mathcal{O}_1, \mathcal{O}_2; \tau) = \text{maximum}\left(\mathcal{O}_1^{\tau}, \mathcal{O}_2^{\tau}\right) \tag{16}$$

where maximum is the operation of taking the maximum value of the element corresponding to the position in the matrices $\mathcal{O}_1^{\tau}$ and $\mathcal{O}_2^{\tau}$. As the model converges, in the later stages of training, the focusing ability of the reference map matures, and an end to the fusion strategy is considered, with only the reference attention map being used to rectify the features

$$\mathcal{V}(\mathcal{O}_1, \mathcal{O}_2; \tau) = \mathcal{O}_1^{\tau}. \tag{17}$$

As a summary, the learning and optimization process of our GMFANet is described in Algorithm 1.



Fig. 6. Some examples of the UCM dataset. Each image corresponds to one of the 21 categories, and the class label is listed below the image.

IV. Experimental Study

A. Datasets, Evaluation Metrics, and Experimental Setup

1) Dataset Details: In order to report the experimental results of our method and all comparison algorithms as well as to validate the performance enhancement of each module of our proposal, three real-world public datasets, including aerial image dataset (AID) [80], NWPU [14], and UC Merced Land-Use (UCM) [81], are selected to implement the benchmark study and analysis. The three benchmark datasets are described in detail as follows.

1) The UCM dataset [81] is a public benchmark dataset for scene classification, which is collected from the United States Geological Survey National Map of the U.S. regions. The 21 scene categories are covered, and each category consists of 100 images with a size of 256 × 256 pixels. Two commonly used data division modes are provided, 50% and 80% of the data for training and the remaining 50% and 20% for testing, respectively. In the presence of rich physical information such as homogeneous or nonhomogeneous texture and color distributions, the UCM dataset plays a nontrivial role for the evaluation of various advanced approaches. Some examples of the UCM dataset are shown in Fig. 6.
2) The AID [80] is a common large aerial imagery dataset for scene classification, released by Wuhan University, Wuhan, China. All images are taken from Google Earth. The AID covers 30 classes and each sample holds a size of 600 × 600 pixels. In addition, a total of

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Fig. 7. Some examples of the AID. Each image corresponds to one of the 30 categories, and the class label is listed below the image.

10 000 images are involved, including two modes of data division, 20% training data and 80% test data, as well as 50% training data and 50% test data. As the data collection is characterized by multiresolution and multisource, it enhances the intraclass diversity of scene images and brings great challenges for scene semantic analysis. Fig. 7 exhibits selected images of AID.

3) The NWPU-RESISC45 dataset (NWPU) [14] is a larger scale scene dataset, i.e., it contains more scene categories, introduced by Northwestern Polytechnical University, Xi'an, China. The 45 category scenes are considered, each of which contains 700 images. Each image is taken from Google Earth and is fixed at 256 × 256 pixels in size. Following the previously mentioned setup of different training modes, the NWPU still has two modes of data division, including 10% and 20% training ratios, with the remaining 90% and 80% used for test. The examples of each category are illustrated in Fig. 8.

2) Evaluation Metrics: Overall accuracy (OA) and confusion matrix (CM) are utilized to quantitatively measure the effectiveness of the method during the experimental implementation. OA is the number of correctly predicted samples divided by the total number of samples to be predicted. CM is employed to discern whether multiple classes are confused, with each column representing the predicted class value and each row representing the actual class. The application of these two indicators allows for the evaluation of the model results at two granularities, the overall performance and the relationship between multiple classes.

3) Experimental Settings: In the experiments, VGG16 and ResNet34 are selected as the backbone for the experiments as

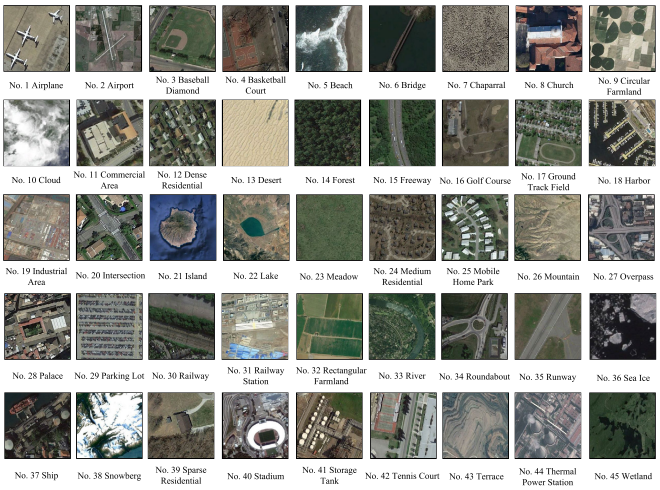


Fig. 8. Some examples of the NWPU dataset. Each image corresponds to one of the 45 categories, and the class label is listed below the image.

a way to validate the generalization of our model. UCM along with 80% training ratio and 20% training ratio, AID with 50% and 20% training ratio, and NWPU together with 20% and 10% partitioning ratio are taken as the six datasets for fair evaluation. In the initialization, the bias parameters and the weights are drawn from a pretrained backbone on ImageNet dataset [92]. There are two critical hyperparameters of our model that need to be specified in advance, i.e., the tradeoff coefficient β between ℓ_{CE} and ℓ_{MSE}, as well as the initial curriculum threshold τ_0. Grid search and cross-validation are performed to search the best value of hyperparameters above. Sensitivity analyses will be conducted in Section IV-C.

In brief, the tradeoff coefficient β is set equally to 0.1 for UCM dataset with 80% ratio training, and AID and NWPU dataset with 10% training ratio division. On the UCM dataset with 50% mode, it is set as 1.0. In addition, β is taken to be 0.01 for NWPU with 20% ratio training. For τ_0, it is set to 0.5 on the UCM dataset and on the other two datasets, AID and NWPU, they take the value of 0.3. In addition, in the experiments, $\Delta\tau$ is set to 0.1 for all datasets for convenience. For input image, data augmentation techniques such as random horizontal flipping, random cropping, and normalization are first employed. SGD is adopted to perform the training with momentum of 0.9 and weight decay of 10^{-4}. Furthermore, models are trained for 120 epochs, and the dynamic learning rate is adopted under the multistep strategy with an initial learning rate of 0.01, milestones of [80, 100], and a regulating factor gamma of 0.1. There is a difference in that in the UCM dataset, batch size is set as 16 for 80% training and 8 for 50% training. In AID, batch size is fixed at 8 for both two training ratios. Furthermore, NWPU prefers batch size with a value of 16 on 20% and 10% training modes. Experiments are all implemented under PyTorch 2.1.2 on NVIDIA RTX 3090 GPU.

B. Comparison With the State-of-the-Art RSSC Models

In order to verify the effectiveness and advanced performance of our proposal, a large number of state-of-the-art methods are chosen as comparison algorithms for experiment. According to the type to which the comparison methods belong, we categorize them into handcrafted feature-based,

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TABLE I

EXPERIMENTAL RESULTS WITH OA ± STANDARD DEVIATIONS (%) OF THE STATE-OF-THE-ART METHODS AND OUR GMFANET ON UCM DATASET, THE BEST PERFORMANCE IS IN BOLD

Type	Methods	Backbone	Publication	OA	
				80% Traning	50% Training
Handcrafted feature-based	BoVW(SIFT) [80]	/	TGRS2017	75.52±2.13	72.40±1.30
	BoVW(LBP) [80]	/	TGRS2017	78.12±1.38	73.48±1.39
	salM^{3}LBP-CLM [82]	/	JSTARS2017	95.75±0.80	94.21±0.75
CNN-based	VGG [69]	VGG16	ICLR2015	96.58±0.41	94.05±0.64
	ResNet [71]	ResNet34	CVPR2016	97.75±0.24	96.05±0.65
	TEX-Net-LF [83]	ResNet50	ISPRS2018	97.72±0.54	96.91±0.36
	ARCNet [59]	VGG16	TGRS2019	99.12±0.40	96.81±0.14
	CAD [84]	DenseNet	JSTARS2020	99.66±0.27	98.57±0.33
	MIDC-Net_CS [85]	DenseNet	TIP2020	97.40±0.48	95.41±0.40
	GCSANet [4]	DenseNet	JSTARS2022	99.31±0.56	98.32±0.71
	SKAL [20]	GoogleNet	TNNLS2022	99.70±0.30	98.19±0.05
	EMTCAL [21]	ResNet34	TGRS2022	99.57±0.28	98.67±0.16
	EMSCNet [86]	ResNet50	TGRS2023	99.44±0.16	98.70±0.46
	EFCOMFF-Net-v1 [87]	ResNet152	TGRS2023	99.85±0.18	99.35±0.10
Transformer-based	TRS [88]	ResNet50+Transformer Encoder	RS2021	99.52±0.17	98.76±0.13
	SCViT [22]	ViT	TGRS2022	99.57±0.31	98.90±0.19
	EMSCNet [86]	ViT-B	TGRS2023	99.71±0.17	99.10±0.20
Ours	GMFANet	VGG16		**99.95±0.08**	**99.88±0.02**
		ResNet34		99.90±0.12	99.63±0.03

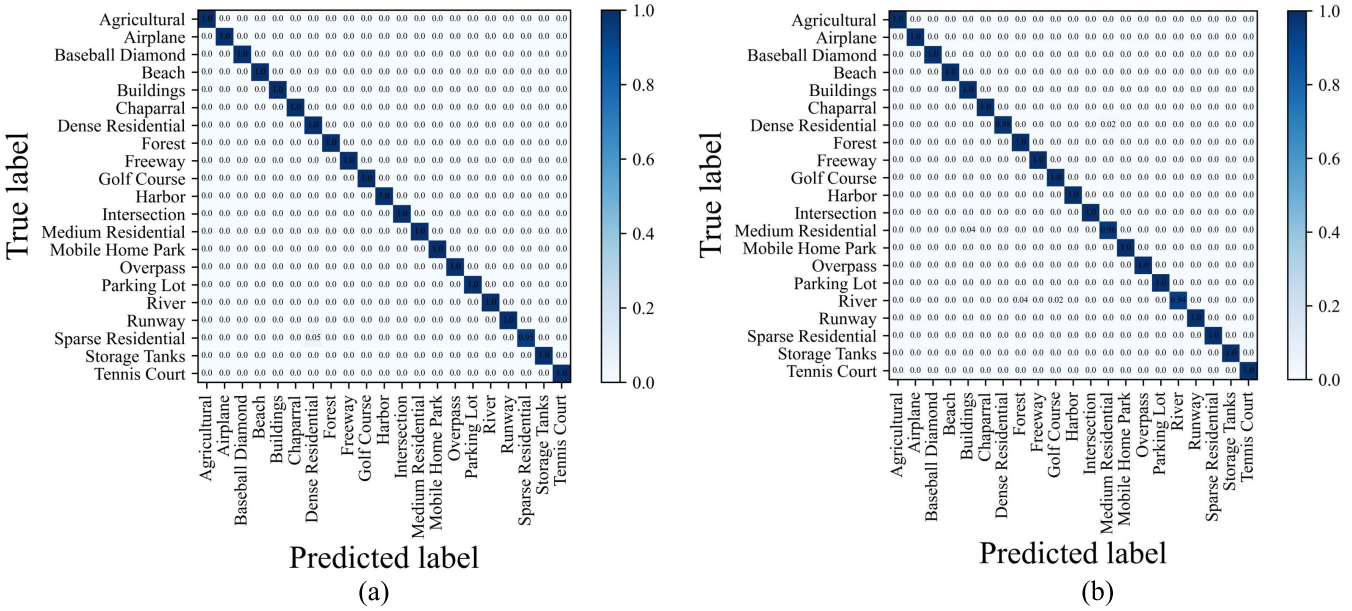


Fig. 9. CMs of GMFANet with ResNet34 backbone on UCM dataset with (a) 80% samples for training and the rest for testing as well as (b) 50% samples for training and the rest for testing.

CNN-based, and transformer-based as a way to reflect the absolute superiority of our GMFANet. The backbone, publication status, OA, and CM performance of GMFANet are reported separately in the results.

1) Experiments on UCM Dataset: Table I shows the results of the comparison methods and our GMFANet on the UCM dataset with 80% training ratio and 50% ratio under repeated five times independently. The best performance is in bold. All the handcrafted feature-based methods are difficult to achieve high OA values. With the introduction of DNNs, TEX-Net-LF is capable of encoding further semantic information based on hand-textured features, which in turn improves the classification performance, but the improvement is not significant compared to the baseline methods (VGG16 and ResNet34) that directly uses raw image input. EFCOMFF-Net-v1 achieves the best performance in the comparison algorithms, with an average OA of 99.85% in the 80% training mode and 99.35% in the 50% training mode. Benefiting from a unique multiscale fusion strategy, it even surpassed the model of the transformer architecture. The transformer-based methods all perform well as a whole, with OA exceeding 99% on an 80% ratio. Unusually, in addition to focusing capability, our model possesses adaptive multiscale fusion inspiration that combines both long-range skip connections and short-range connections. It can be found that our GMFANet outperforms all the comparison methods and achieves the best performance on OA both in terms of mean and standard deviation. The OA values are 99.95% and 99.88%, respectively. According to the

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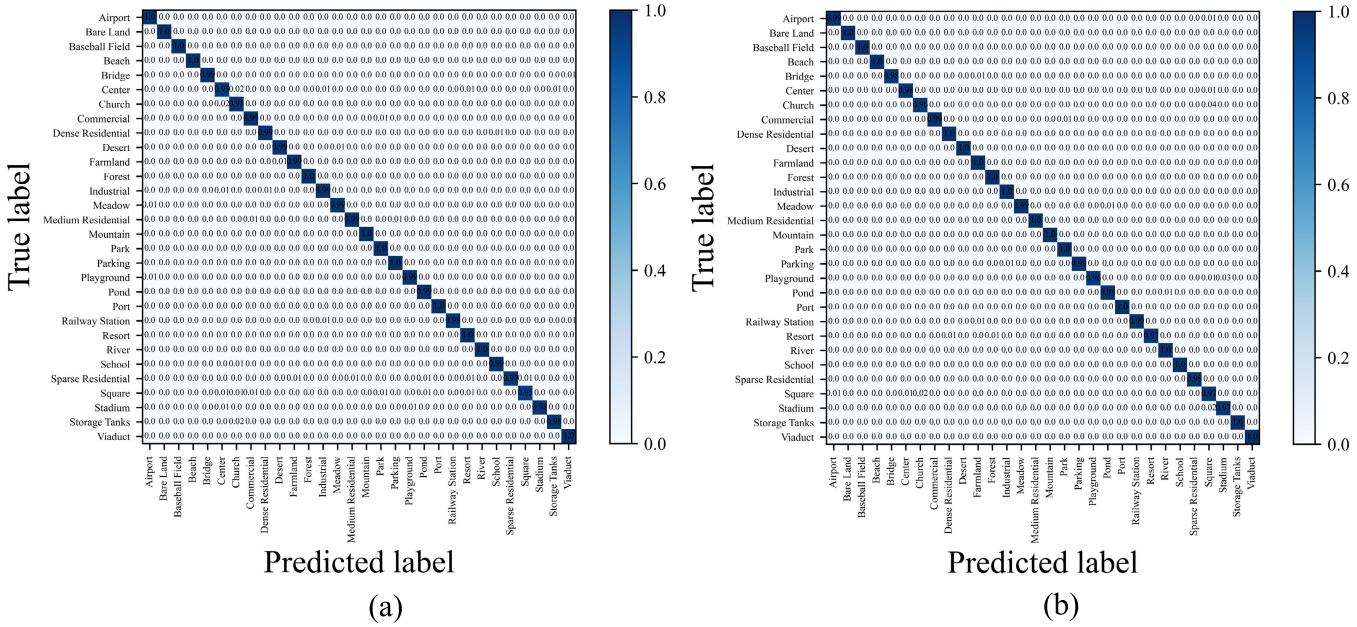


Fig. 10. CMs of GMFANet with ResNet34 backbone on AID with (a) 50% samples for training and the rest for testing as well as (b) 20% samples for training and the rest for testing.

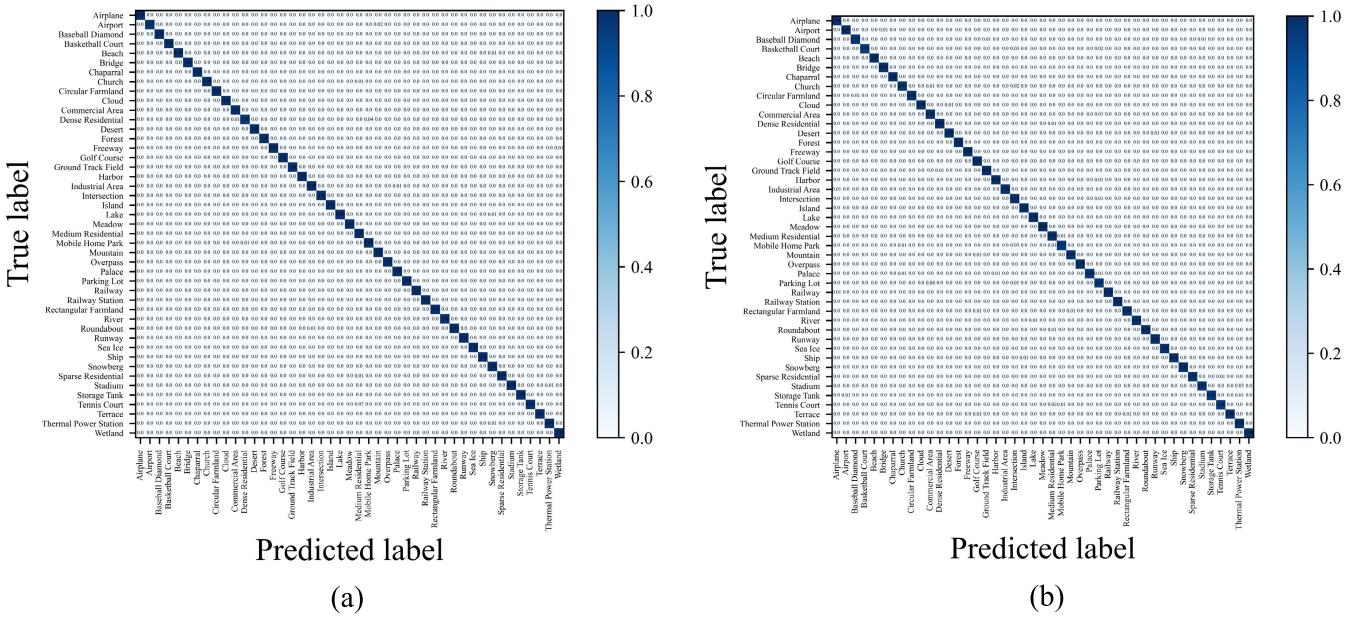


Fig. 11. CMs of GMFANet with ResNet34 backbone on NWPU dataset with (a) 20% samples for training and the rest for testing as well as (b) 10% samples for training and the rest for testing.

performance on OA above, it almost reaches the upper limit of performance on both ratio divisions, which is close to 100% classification accuracy. The lowest OA standard deviation also reflects the excellent stability of the GMFANet.

In order to have a clearer view of the performance of each category, the CM of GMFANet is plotted in Fig. 9. It can be seen that the accuracy of all the categories, except category "Sparse Residential," reaches 100% for 80% proportion of training data. As for the 50% division pattern, the accuracy of scene recognition exceeds 95% for all categories except for the "river" category. Among them, the accuracy reaches 98% for the category "Dense Residential" and 96% for the category "Medium Residential." It can be noticed that residential houses are easily confused with building-intensive scenes since they contain a large number of building structures.

2) Experiments on AID: The experimental results on the AID with 50% and 20% ratio training sets, respectively, are reported in Table II. None of the approaches based on handcrafted features exceeded 90% on OA, where salM^{3}LBP-CLM has a more prominent performance, but only obtained 89.76% and 86.92%. The CNN-based and transformer-based approaches achieve consistently competitive performances. Apparently, the comparison algorithms, such as CAD, GCSANet, and LSCNet, based on the traditional

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TABLE II
EXPERIMENTAL RESULTS WITH OA ± STANDARD DEVIATIONS (%) OF THE STATE-OF-THE-ART METHODS AND OUR GMFANET ON AID, THE BEST PERFORMANCE IS IN BOLD

Type	Methods	Backbone	Publication	OA	
				50% Traning	20% Training
Handcrafted feature-based	BoVW(SIFT) [80]	/	TGRS2017	68.37±0.40	62.49±0.53
	BoVW(LBP) [80]	/	TGRS2017	64.31±0.41	56.98±0.55
	salM^{3}LBP-CLM [82]	/	JSTARS2017	89.76±0.45	86.92±0.35
CNN-based	VGG [69]	VGG16	ICLR2015	95.37±0.09	92.76±0.10
	ResNet [71]	ResNet34	CVPR2016	94.12±0.19	93.23±0.12
	TEX-Net-LF [83]	ResNet50	ISPRS2018	95.73±0.16	93.81±0.12
	ARCNet [59]	VGG16	TGRS2019	93.10±0.55	88.75±0.40
	CAD [84]	DenseNet	JSTARS2020	97.16±0.26	95.73±0.22
	MIDC-Net_CS [85]	DenseNet	TIP2020	92.95±0.17	88.51±0.41
	SCCov [19]	VGG16	TNNLS2020	96.10±0.16	93.12±0.25
	GCSANet [4]	DenseNet	JSTARS2022	97.53±0.32	95.96±0.38
	SKAL [20]	ResNet18	TNNLS2022	96.76±0.20	94.38±0.10
	EMTCAL [21]	ResNet34	TGRS2022	96.41±0.23	94.69±0.14
	EMSCNet [86]	ResNet50	TGRS2023	96.96±0.10	95.13±0.10
	LSCNet [89]	Convolutional Block	TGRS2023	97.14±0.14	95.38±0.15
	EFCOMFF-Net-v1 [87]	ResNet152	TGRS2023	98.70±0.07	96.44±0.07
	MGSNet [28]	Convolutional Block	TGRS2023	97.18±0.16	95.46±0.21
	CGINet [90]	ResNet50	TGRS2024	97.10±0.24	95.35±0.14
Transformer-based	TRS [88]	ResNet50+Transformer Encoder	RS2021	98.48±0.06	95.54±0.18
	SCViT [22]	ViT	TGRS2022	96.98±0.16	95.56±0.17
	EMSCNet [86]	ViT-B	TGRS2023	97.35±0.17	96.02±0.18
Ours	GMFANet	VGG16		99.07±0.10	**99.97±0.05**
		ResNet34		**99.73±0.09**	98.80±0.41

attention mechanism, the comparison algorithms, CAD, SCCov, and GCSANet, based on dense connections, and SKAL and MGSNet based on the selection of key landmarks are all capable of delivering not-so-low gains. EFCOMFF-Net-v1 obtains the best OA on both proportions of training data, 98.70% and 96.44%, respectively, and also outperforms the model with the transformer backbone. In addition, on 50% training, due to the unique hybrid architecture of convolution and transformer, TRS with ResNet50 and transformer encoder harvests the best result 98.48% in transformer-based models, second only to EFCOMFF-Net-v1. GMFANet outperforms all comparative algorithms, reaching 99.73% at 50% and 99.97% at 20% training ratio. It is 1.03% and 3.53% higher than the second-best method EFCOMFF-Net-v1. It is easy to see that our proposal has a significant competitive advantage and is more profitable.

The CMs of GMFANet on AID with 50% and 20% training ratios are depicted, as shown in Fig. 10. All of the category scenarios predictions are over 95%. For the 50% training ratio, 12 of the 30 categories reach 100%. As can be seen, compared to the UCM dataset, more confusion does occur as the number of categories increases, e.g., "Sparse Residential," "Square," and "Censer." It can be seen that elements containing sparse buildings appear more likely to be misclassified. On the 20% training ratio, the lowest accuracy belongs to "Playground" and "Church" which reaches 96%. It can be observed that "Church" is easily recognized as "Square" and "Playground" is more easily confused with "Stadium." Indeed, the decreasing training data cause a certain degree of performance decay; however, a comparison with other state-of-the-art approaches reveals that this effect is limited with the enchantment of our proposal.

3) Experiments on NWPU Dataset: The NWPU dataset possesses more scene categories, which presents a challenge for scene semantic analysis. The 20% and 10% training

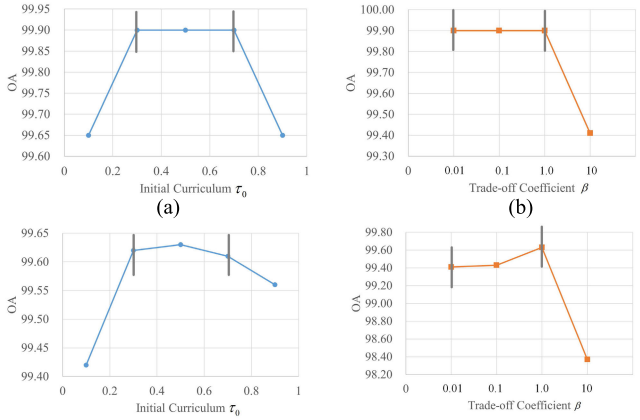


Fig. 12. Sensitivity analyses of GMFANet with ResNet34 backbone on UCM dataset with 80% and 50% training ratio. (a) Initial curriculum on UCM dataset with 80% training ratio. (b) Tradeoff coefficient on UCM dataset with 80% training ratio. (c) Initial curriculum on UCM dataset with 50% training ratio. (d) Tradeoff coefficient on UCM dataset with 50% training ratio.

divisions are set to implement the experiments, and the results are reported in Table III. It can be evident that our GMFANet with both two backbones achieves leading accuracy on this dataset. It obtains 98.61% OA at 20% ratio, outperforming the second-best method EFCOMFF-Net-v1 in comparison approaches by 2.73%. In the comparison algorithms, only EFCOMFF-Net-v1, TRS, and EMSCNet exceed 95%. Compared to the baseline (VGG or ResNet), the respective introduction of multiscale learning and attention mechanisms has improved the model's semantic understanding of critical geographical objects, and as a result, comparison algorithms such as EMSCNet, LSCNet, and EFCOMFF-Net-v1 have also achieved competitive performances. However, the transformer architecture with global awareness has a better performance on this complex dataset. For the 10% training mode, the reduction in training data further puts a damper

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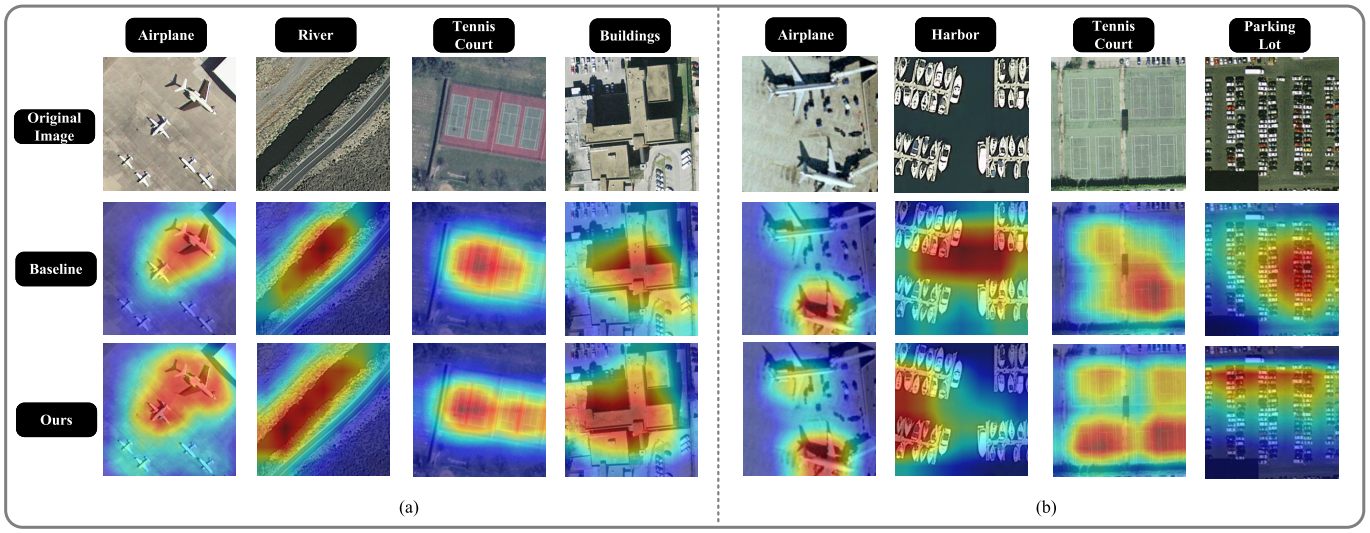


Fig. 13. Heat maps of the visualization results for ResNet-34 baseline and GMFANet on the UCM dataset with (a) 80% and (b) 50% training ratios, the first row shows the original input image, the second row shows the visualization results of the baseline, and the third row shows the visualization results of our proposal.

TABLE III

EXPERIMENTAL RESULTS WITH OA ± STANDARD DEVIATIONS (%) OF THE STATE-OF-THE-ART METHODS AND OUR GMFANET ON NWPU DATASET, THE BEST PERFORMANCE IS IN BOLD

Type	Methods	Backbone	Publication	OA	
				20% Traning	10% Training
Handcrafted feature-based	BoVW [14]	/	RPOC2017	44.97±0.28	41.72±0.21
	BoVW+SPM [14]	/	RPOC2017	32.96±0.47	27.83±0.61
	LLC [14]	/	RPOC2017	40.03±0.34	38.81±0.23
CNN-based	VGG [69]	VGG16	ICLR2015	90.64±0.14	87.14±0.17
	ResNet [71]	ResNet34	CVPR2016	92.12±0.09	89.20±0.14
	CAD [84]	DenseNet	JSTARS2020	94.58±0.26	92.70±0.32
	MIDC-Net_CS [85]	DenseNet	TIP2020	87.99±0.18	86.12±0.29
	SCCov [19]	VGG16	TNNLS2020	92.10±0.25	89.30±0.35
	GCSANet [4]	DenseNet	JSTARS2022	93.95±0.36	93.39±0.39
	SKAL [20]	GoogleNet	TNNLS2022	92.95±0.09	90.41±0.12
	EMTCAL [21]	ResNet34	TGRS2022	93.65±0.12	91.63±0.19
	T-CNN [91]	ResNet18	TGRS2022	93.05±0.12	90.25±0.14
	MBFANet [45]	ResNet50	GRSL2023	94.01±0.08	91.61±0.14
	EMSCNet [86]	ResNet50	TGRS2023	94.08±0.20	92.16±0.07
	LSCNet [89]	Convolutional Block	TGRS2023	94.54±0.19	92.80±0.14
	EFCOMFF-Net-v1 [87]	ResNet152	TGRS2023	95.88±0.11	93.62±0.14
	MGSNet [28]	Convolutional Block	TGRS2023	94.57±0.12	92.40±0.16
	CGINet [90]	ResNet50	TGRS2024	94.38±0.13	92.28±0.17
Transformer-based	TRS [88]	ResNet50+Transformer Encoder	RS2021	95.56±0.20	93.06±0.11
	SCViT [22]	ViT	TGRS2022	94.66±0.10	92.72±0.04
	EMSCNet [86]	ViT-B	TGRS2023	95.37±0.07	93.58±0.22
Ours	GMFANet	VGG16		98.13±0.24	97.65±0.32
		ResNet34		**98.61±0.56**	**97.66±0.67**

on model performance. No state-of-the-art comparison method exceeds 94%; however, GMFANet reaches 97.66%, which is 4.04% higher than the second-best method, EFCOMFF-Net-v1. It can be concluded that our approach can greatly improve the scene analysis capability of deep models under the robustness to variable-scale information and accurate perception as well as localization of critical geographical objects, both compared to CNN-based methods and transformer-based methods.

In order to better analyze the effect of GMFANet on the confusion between categories, the CM under ResNet34 backbone is further plotted, as shown in Fig. 11. CM of GMFANet with 20% training ratio indicates that only "Airport" and "Dense Residential" have a slightly higher degree of confusion, tending to be predicted as "Mountain" and "Mobile Home Park," respectively. All other scenario categories have prediction accuracies exceeding 98%. For CM on 10% training ratio, "Basketball Court" is easier to predict as "Intersection" and "Parking Lot." "Mobile Home Park" is also easily confused to class "Intersection," while some "Parking Lot" cases might be misclassified as "Commercial Area." Other than that, the prediction accuracy of all other categories of scenarios exceeds 95%.

C. Sensitivity Analyses

In this section, we perform a sensitivity analysis on the critical hyperparameters, the tradeoff coefficient β between ℓ_{CE} and ℓ_{MSE}, as well as the initial curriculum threshold τ_0,

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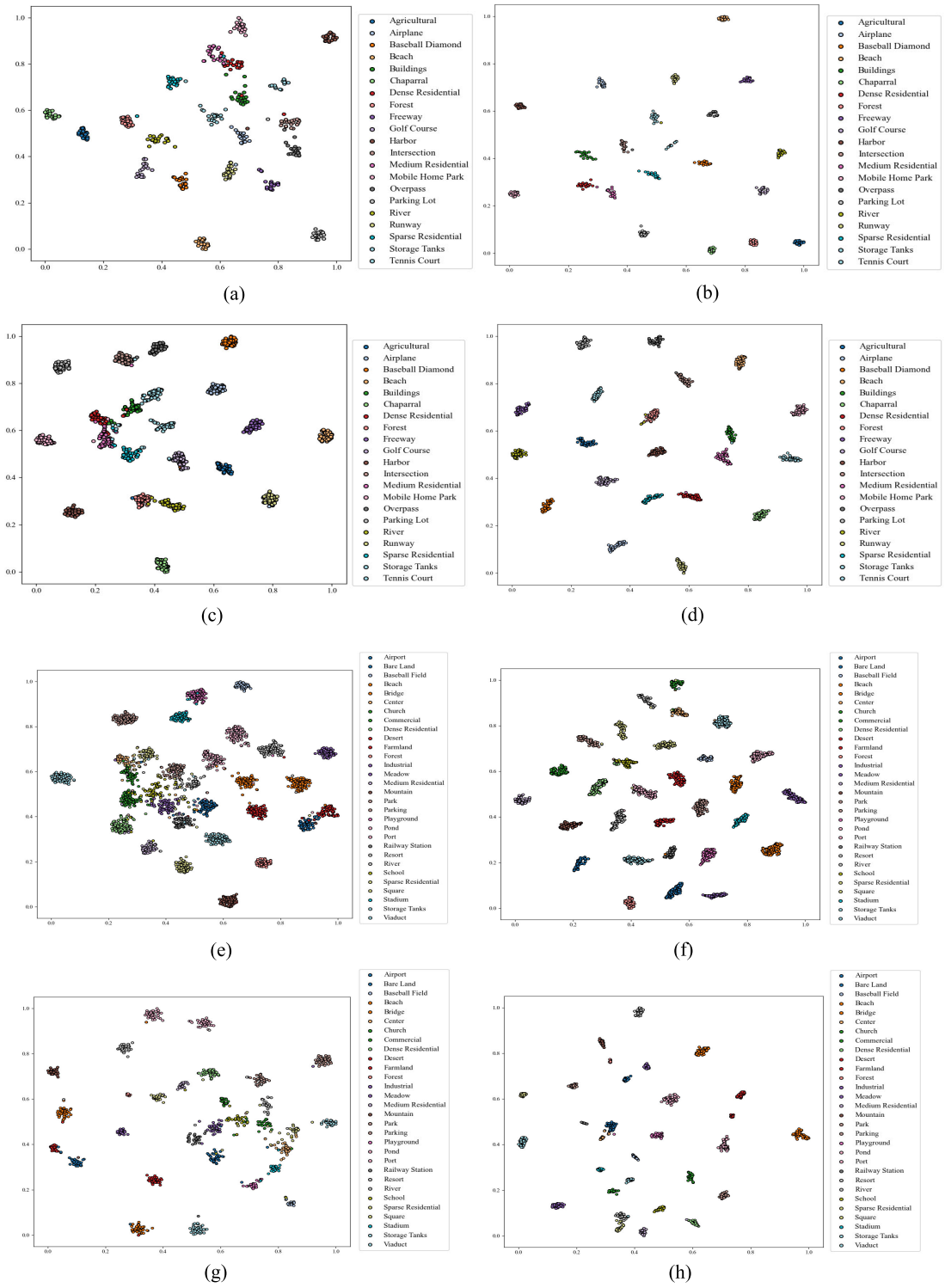


Fig. 14. t-SNE visualization results of (a) baseline and (b) GMFANet on UCM dataset with 80% training ratio, (c) baseline and (d) GMFANet on UCM dataset with 50% training ratio, (e) baseline and (f) GMFANet on AID with 50% training ratio, and (g) baseline and (h) GMFANet on AID with 20% training ratio.

involved in GMFANet. Grid search and cross-validation are adopted to search the best values in the experiment.

The initial curriculum threshold τ_0 determines the starting point in the process of fusion from easy to hard for the attention map, and it will have an impact on the model performance. The results of the sensitivity analysis with ResNet34 backbone on the UCM dataset with 80% and 50% training ratios are shown in Fig. 12(a) and (c). The β is fixed at 0.1 to search for the initial curriculum threshold τ_0, and we fine-tune τ_0 in the interval [0.1, 0.9] with a step size of 0.2. It can be seen that the parameter variations on both datasets are in a relatively smooth state between 0.3 and 0.7 with a maximum value achieved. In the 80% mode, the performance is identical, with 99.90% achieved, while in the 50% mode, τ_0 peaks at 0.5 for OA,

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TABLE IV
ABLATION ANALYSIS OF GMFANET UNDER DIFFERENT BACKBONES ON UCM DATASET. "√" INDICATES THE MODULE CONTAINED, AND "×" INDICATES THE MODULE NOT CONTAINED. THE BEST PERFORMANCE IS IN BOLD

Backbone	Baseline	w/ HMA	w/ GSFA	w/ CDAF	OA	
					80% Traning	50% Training
VGG16	√	×	×	×	96.58	94.05
	√	√	×	×	98.31	96.46
	√	×	√	×	99.61	99.58
	√	×	√	√	99.82	99.74
	√	√	√	√	**99.95**	**99.88**
ResNet34	√	×	×	×	97.75	96.05
	√	√	×	×	98.13	96.98
	√	×	√	×	99.56	99.01
	√	×	√	√	99.76	99.14
	√	√	√	√	**99.90**	**99.63**

i.e., 99.63%. Based on this experience, τ_0 on the AID and NWPU dataset can also be reduced to [0.3, 0.7] for the test. Ultimately, 0.3 is suggested.

The tradeoff coefficient β determines the involvement of the auxiliary task ℓ_{MSE}, which is utilized to subtly steer the learning direction of the attention generating network. Due to the presence of classification loss, the prediction attention map already has the ability to automatically learn focal attention, which is considered as a complement to the reference attention map. Therefore, the hyperparameter β is favored to be set to a smaller value. With the fixed τ_0 above, the results of the sensitivity analysis on the UCM dataset with 80% and 50% training ratios are shown in Fig. 12(b) and (d). We test on the tuning set [0.01, 0.01, 0.1, 1.0, 10.0]. Results in the 0.01–0.1 range are more acceptable. For 80% mode, it can achieve 99.90% accuracy simultaneously. However, on the 50% ratio training, β takes 1.0 to have the best performance, with an OA of 99.63%. In addition, we set β to 0.1 on the AID, and β is set to 0.01 on the NWPU dataset with 20% ratio training and set to 0.1 on the 10% ratio training.

D. Ablation Study

In order to further validate the enhancement of the baseline by our GMFANet and the important role of the proposed individual modules, in this section, an ablation analysis is carried out. Table IV lists the ablation results of GMFANet under VGG16 and ResNet34 backbones on the UCM dataset. It is worth noting that GSFA and CDAF are tightly coupled, i.e., CDAF is a learning strategy for the attention fused in GSFA. Therefore, there is no way to test CDAF alone in the absence of GSFA. It can be noticed that with the introduction of each module, either HMA, GSFA, or CDAF, there is a positive effect on performance improvement. Among them, the effect of GSFA is more obvious, which improves the accuracy by 3.03% for baseline with VGG16 backbone under 80% training, and by 5.53% under 50% training. The model achieves the best results when HMA, GSFA, and CDAF are combined together under both VGG16 and ResNet34 backbones.

As a result of the extensive quantitative analysis described above, it has been concluded that GMFANet can achieve the state-of-the-art performance on several public datasets and is close to the upper performance limit of the data. In order to demonstrate more intuitively the multiscale focusing capability of our proposal, Fig. 13 shows the visualization heat maps of baseline and GMFANet with ResNet34 backbone on some instances of UCM dataset under different training ratios. It can be found that at an 80% training ratio, GMFANet can pay extra attention to small-scale airplanes in the "Airplane" scenario compared to the baseline. Through the "Tennis Court" examples at 80% and 50% ratios, the ability of our model to focus on critical geographical regions is significantly improved.

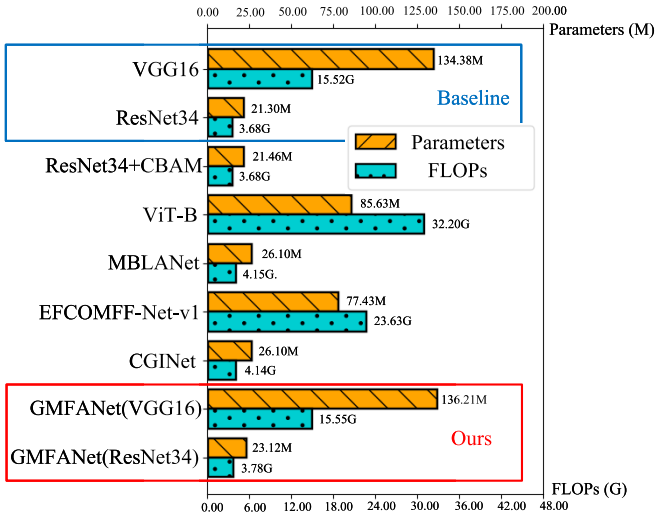


Fig. 15. Parameters (M) and FLOPs (G) of baselines, comparison methods, and our GMFANet.

In addition, we plot the t-distributed stochastic neighbor embedding (t-SNE) visualization [93] results of ResNet34 and GMFANet on the UCM dataset and AID in Fig. 14. On all data, our GMFANet can greatly enhance the distinguishability of geographical representation compared to the baseline. The similarity between classes decreases tremendously and the similarity within classes increases extremely. For example, the categories "Dense Residential," "Medium Residential," and "Buildings" are prone to overlap for baselines on the UCM dataset with a 50% training ratio, while GMFANet greatly solves this problem.

In order to better explain the computational complexity and the number of parameters (Parameters) of our model, Fig. 15 shows the floating-point operations (FLOPs) and Parameters of our method under two backbones (VGG16 and ResNet34), baselines (VGG16 and ResNet34), and comparison algorithms (ResNet34 + CBAM [25], ViT-B [58], MBLANet [94], EFCOMFF-Net-v1 [87], and CGINet [90]). Compared to baseline, our GMFANet brings only about 2M parameters count increase and no more than 0.1G increase in FLOPs. The low computational complexity and parameter rise are attributed to our lightweight parameterized hierarchical multiscale attention mechanism. Even though CBAM can bring about a fewer increase in parameters and computational complexity, it has inherent perceptual bottlenecks in scene classification, as depicted in Fig. 1. GMFANet, with robust multiscale adaptive fusion and spatial focusing capabilities, can bring about better performance gains along with a slight increase in both parameters and FLOPs.

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V. Conclusion

In order to solve the two major difficulties encountered in the analysis of remote sensing image scenes, i.e., variable-scale targets and the hard spotlighting on critical geographical objects separated from the background, in this article, we specifically propose the GMFANet for RSSC to adaptively localize the representative multiscale semantic representation for complex scenes. In particular, we present the HMA scheme, GSFA mechanism, and CDAF strategy as the key components of the GMFANet, which can adaptively augment physical details and high-level semantic information, accurately locate critical regions separated from the background, and suppress learning ambiguities during optimization. Extensive comparative experiments, ablation studies, and visualization analyses have shown that GMFANet contributes significantly to the extraction of distinguishable and representative geographical semantic features.

However, the method in this article is still constrained by the accuracy of gradient guidance network in generating reference map in the presence of noise or interference attacks, and we plan to focus our next work on a more robust and generalized focal attention generation. In future research, we will also focus more on exploring the semantic analysis of scenes with few-shot setting or even in unsupervised RSSC tasks.

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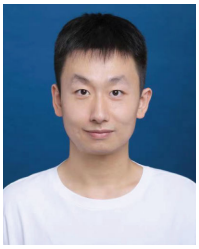
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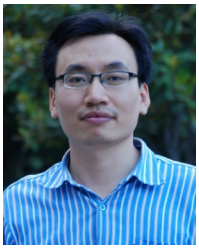
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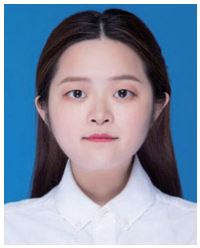
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